

# Comparison of Clinical Outcome of Periapical Surgery in Endodontic and Oral Surgery Units of a Teaching Dental Hospital

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Outcome of Periapical Surgery in Endodontic and Oral Surgery

## INTRODUCTION

Apical periodontitis (AP) or periapical periodontitis is an inflammatory lesion around the apex of a tooth root usually caused by invasion of microorganisms (Bacteria) in tooth pulp<sup>1</sup>. Some dentists thought that periapical disease can be managed with root canal method because of its high success rate about 98%<sup>2</sup>. But in cases in which root canal fails incidence of failure must be kept in mind before start of further management strategy. Causes of failure include resistant intracanal infection, coronal leakage, extra radicular infection, cyst, by residual intracanal infection and foreign body reaction<sup>3</sup>.

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Sometime lesions present in mandible or maxillary bone or around its several roots which may destroy the support of tooth and cause infection and moderate to severe pain<sup>4</sup>. Size of lesion vary from small (< centimeter) that may be treated with root canal of the tooth by a senior odontostomatologist<sup>5</sup>. In most of cases root canal solve the problem of patients but not all cases. In some cases when root canal doesn't resolve the lesion repeat of root canal is suggested<sup>6</sup>. Periapical surgery will be indicated if second attempt of root canal is failed. These lesions are called granulomas and periapical cysts and its origin it's in a chronic dental infection<sup>7</sup>.

Surgical extraction of tooth root is the main part of periapical surgery which removes the lesion thoroughly<sup>8</sup>. This procedure usually accompanied by preparation of root cutting of tooth or sealing off with special cement or amalgam<sup>9</sup>. If periapical lesion is not removed properly or treated aggressively it can cause multiple infections or increase in lesion size, infection of adjuvant teeth and destruction of bone<sup>10</sup>.

Only two options are alternatives if periapical lesion hasn't respond to root canal procedure<sup>11</sup>. One is periapical surgery and 2<sup>nd</sup> is exodontias of tooth. Other than periapical surgery exodontias have advantage of early healing and disadvantage of tooth losing which may be restored with prosthetic later on. On other hand

periapical surgery have advantage of tooth keeping and disadvantage of slow and lesser healing rate<sup>11</sup>.

Conservative treatment always appreciated and regarded as best treatment choice<sup>12</sup>. But a case in which further treatment with conservative method is not possible periapical surgery is an ideal alternative treatment<sup>13</sup>. Periapical surgery may be treatment of choice if conservative treatment gives poor outcomes. Success rate of periapical surgery is about 95% that varies case to case and depends upon treatment procedure, case selection, statistical analysis, evaluation period and most likely criteria of success evaluation<sup>14</sup>.

Many studies have been conducted to evaluate the effect of different variables on outcomes of periapical surgery and all authors develop consensus that sex, age, tooth type preoperative signs statistically affect the surgical healing after operative procedure and other influencing factors are contradictory<sup>15</sup>. It was also reported that presence of periapical radiolucency before surgery affects the surgical outcomes after periapical surgery. Quality and composition of root canal filling is a climacteric factor but some controversial studiers also found. Some authors support material in place of amalgam<sup>16</sup>.

Oral surgeons and endodontists both perform surgery of periapical region but their clinical skill, approaches, culture, philosophies, training pathway and attitudes are different that may affect outcomes significantly<sup>1</sup>.

## MATERIALS AND METHODS

Patients who had periapical surgery at Bibi Aseefa Dental College, Larkana were enrolled in study. Study was started after ethical approval from hospital ethical board. Informed written consent was obtained from patients and they were ensured about their confidentiality. Non probability consecutive sampling technique was used.

A total of 120 patients was included in study and divided into two groups (Group A and B) 60 patients in each group. Patients in group A were operated by endodontist and in group B were operated by oral surgeon. All types of tooth were included. All patients were assessed in endodontic unit before periapical surgery. Prerequisite of surgery were coronal restoration and satisfactory root filling. High speed hand pieces, ultrasonic retro tips amalgam was used in all cases. During surgical procedure radiographs were taken to check the status and placement of material in periradicular tissues. Postoperative analgesics, antibiotics and mouthwash were advised. After one week sutures were removed and patients were followed up till 2 years biannually.

Treatment protocol in oral surgery department was not different but Operative procedure was done by using slow speed hand piece, tungsten carbide burs and amalgam was used. Intraoperative radiographs and post-operative analgesics, antibiotics and mouth wash

were advised. Sutures removed after 14 days. Two years follow up was completed.

Outcomes measures were assessed radiographically and clinically and recorded on a predesigned performa. Data

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## CONCLUSION

There is no significant difference regarding radiographic and clinical success and failure between periapical surgery in endodontic and oral surgery units. Quality of filling and filling material are two main contributing factors of periapical surgery.

### Author's Contribution:

Concept & Design of Study: Nadia Bashir  
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